

House Price Prediction System: Model Optimization & Performance Comparison

Project: Artificial Intelligence & Machine Learning - Task 2

1. Objective

The objective of this project is to build an enhanced Machine Learning system to predict house prices by applying industry-aligned feature engineering and optimization techniques [cite: 1]. This involves preparing the data correctly through scaling, training multiple regression algorithms, evaluating their performance using measurable metrics, and selecting the best-performing model based on the results [cite: 1].

2. Methodology

The workflow followed a structured Machine Learning pipeline designed to ensure reliable and fair model evaluation:

- **Dataset:** The project utilizes the official California Housing Dataset to predict the target variable, *Median House Value* (HousePrice), based on input features such as median income, house age, average rooms, and population [cite: 1, 2].
- **Data Separation:** The dataset was separated into input features (X) and the target variable (y) to prepare for supervised learning [cite: 1, 2].
- **Feature Scaling:** Because the input features exist on vastly different numeric ranges, StandardScaler was applied to the feature set (X) [cite: 1, 2]. This is a critical preprocessing step that gives all features a common scale, preventing certain features from dominating the learning process and ensuring stable model performance [cite: 1].
- **Train-Test Split:** The scaled data was split using an 80/20 ratio (test_size=0.2, random_state=42) [cite: 2]. This split guarantees that the models are evaluated on unseen data, testing their true predictive power rather than memorized values [cite: 1].
- **Model Selection:** Instead of relying on a single algorithm, three distinct regression models were trained for comparison [cite: 1]:
 1. **Linear Regression:** Trained as the baseline predictive model [cite: 1, 2].
 2. **Ridge Regression:** Configured with alpha=1.0 to introduce regularization and reduce potential overfitting [cite: 1, 2].
 3. **Decision Tree Regressor:** Configured with a max_depth=5 to capture non-linear relationships within the housing data [cite: 1, 2].

3. Results & Performance Comparison

The models were evaluated using two primary metrics: Root Mean Squared Error (RMSE) and the R^2 Score [cite: 1, 2]. A lower RMSE indicates superior prediction accuracy, while a higher R^2 score represents better explanatory power [cite: 1]. The comparative evaluation yielded the following results:

Model	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

4. Conclusions & Model Selection

Based on the objective evaluation metrics, the **Decision Tree Regressor** is the best-performing model for this dataset.

- **Accuracy:** The Decision Tree achieved the lowest Root Mean Squared Error (0.724234), meaning its predictions deviated the least from the actual house prices [cite: 2].
- **Explanatory Power:** It also achieved the highest R^2 Score (0.599732), indicating that it successfully captured nearly 60% of the variance in the target variable, outperforming the linear models [cite: 2].

While the Linear Regression and Ridge Regression models performed nearly identically—with Ridge offering a negligible mathematical improvement over the baseline—both were outperformed by the Decision Tree [cite: 2]. This suggests that the relationship between the California housing features (like location, income, and age) and the median house value contains non-linear patterns that the Decision Tree was better equipped to learn [cite: 1, 2]. Therefore, the Decision Tree Regressor is the selected model for this prediction system.